## Presentation Speech

**[Slide 1 - Autocorrelation & Volatility Clustering]**

In this phase of the analysis, we focused on the statistical properties of returns to guide our choice of optimal replication models.

By analyzing autocorrelation through ACF, PACF, and Ljung-Box tests, we observed that returns behave primarily as 'White Noise'. This means past returns offer no direct linear predictive power to estimate future movements.

However, the situation changes when we analyze risk. By applying Engle's ARCH test on squared returns, we found severe conditional heteroskedasticity, with a p-value close to zero. In other words, there is a strong phenomenon of 'volatility clustering': markets alternate between calm periods and periods where shocks cluster together.

These statistical findings have two clear implications for our model:

1. **For tracking:** The lack of serial correlation prompts us to discard static models, requiring advanced and dynamic architectures.
2. **For risk control:** The predictability of volatility spikes mathematically justifies our EWMA VaR Scaling to automatically deleverage the portfolio during crises.

**[Slide Transition 2 - Target Risk Profile & Scatter Plot]**

Moving on to the risk profile of our Synthetic Target, we can see that at first glance, the annualized average volatility seems contained, but this is a deceptive metric.The target, in fact, suffers from heavy left tails, high kurtosis, and severe drawdowns recorded during major market crashes, such as in 2008 and 2020.

If we look at the risk-return scatter plot, we can observe that the target is positioned in a region characterized by low volatility. Conversely, individual futures carry substantially higher intrinsic risk. This presents us with a clear 'scaling challenge': a 1-to-1 replication using these futures is impossible because it would overshoot our risk budget. To reproduce both returns and downside sensitivity, the clone requires multi-asset diversification and algorithmic deleveraging.

**[Slide Transition 3 - Rolling Dynamics & Single-Factor Limits]**

Finally, we extended our analysis by moving from a static to a dynamic view, calculating rolling metrics over a 52-week window. This step is crucial because 'full-sample' metrics calculated over the entire period can be misleading: market regimes change, and with them, both correlations and betas fluctuate dramatically.

From this dynamic analysis, it emerges that the S&P 500 future is the most stable systematic driver for our target, recording the highest average correlation. However, even using the best possible driver, we encounter what we have defined as the 'Single-Factor Limit'. As can be seen from the beta-adjusted Tracking Error chart, the tracking error of a single future remains at unacceptably high levels, and commodities introduce excessive noise.

Our final modeling choice is that no single contract can replicate the complexity of the target. A dynamic, multi-factor replica is strictly required.